

VESSEL ANALYSIS REPORT

vaf(%)	branch points count	tip point count	vein count	total vein length	average vein length
2.578	135.0	136.0	271.0	7517.196	27.739

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
1	[112,0]	[95,6]	18.485	TIP	BRANCH
2	[95,6]	[86,45]	41.728	BRANCH	BRANCH
3	[70,0]	[95,6]	26.071	TIP	BRANCH
4	[406,9]	[371,47]	63.213	BRANCH	BRANCH
5	[406,9]	[438,14]	34.314	BRANCH	BRANCH
6	[406,9]	[418,30]	25.142	BRANCH	BRANCH
7	[138,8]	[137,11]	2.414	TIP	BRANCH
8	[158,19]	[137,11]	23.314	TIP	BRANCH
9	[131,16]	[137,11]	6.657	TIP	BRANCH
10	[438,14]	[460,23]	25.314	BRANCH	BRANCH
11	[438,14]	[418,30]	25.213	BRANCH	BRANCH
12	[460,23]	[473,81]	108.527	BRANCH	BRANCH
13	[460,23]	[450,69]	48.728	BRANCH	BRANCH
14	[62,0]	[54,24]	26.314	TIP	BRANCH
15	[0,25]	[54,24]	71.113	TIP	BRANCH
16	[54,24]	[66,45]	24.971	BRANCH	BRANCH
17	[418,30]	[415,51]	21.243	BRANCH	BRANCH
18	[342,17]	[302,32]	45.799	TIP	BRANCH
19	[260,39]	[302,32]	44.728	TIP	BRANCH
20	[291,54]	[302,32]	28.485	TIP	BRANCH
21	[26,30]	[15,40]	15.314	TIP	BRANCH
22	[0,34]	[15,40]	17.657	TIP	BRANCH
23	[6,56]	[15,40]	22.485	TIP	BRANCH
24	[66,45]	[86,45]	19.828	BRANCH	BRANCH
25	[66,45]	[65,71]	25.414	BRANCH	BRANCH
26	[86,45]	[88,66]	20.828	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
27	[245,3]	[238,45]	43.899	TIP	BRANCH
28	[222,50]	[238,45]	17.071	TIP	BRANCH
29	[242,53]	[238,45]	8.243	TIP	BRANCH
30	[371,47]	[346,70]	37.627	BRANCH	BRANCH
31	[371,47]	[384,64]	21.556	BRANCH	BRANCH
32	[415,51]	[390,64]	29.385	BRANCH	BRANCH
33	[415,51]	[450,69]	41.042	BRANCH	BRANCH
34	[384,64]	[390,64]	5.0	BRANCH	BRANCH
35	[384,64]	[374,77]	15.728	BRANCH	BRANCH
36	[390,64]	[401,83]	22.142	BRANCH	BRANCH
37	[118,85]	[88,66]	39.456	TIP	BRANCH
38	[88,66]	[65,71]	24.071	BRANCH	BRANCH
39	[450,69]	[451,79]	9.414	BRANCH	BRANCH
40	[341,72]	[346,70]	4.828	TIP	BRANCH
41	[346,70]	[354,78]	9.899	BRANCH	BRANCH
42	[65,71]	[17,117]	68.983	BRANCH	BRANCH
43	[374,77]	[354,78]	19.414	BRANCH	BRANCH
44	[374,77]	[389,103]	30.799	BRANCH	BRANCH
45	[350,85]	[354,78]	7.657	TIP	BRANCH
46	[451,79]	[473,81]	21.828	BRANCH	BRANCH
47	[451,79]	[443,89]	11.899	BRANCH	BRANCH
48	[473,81]	[488,119]	42.799	BRANCH	BRANCH
49	[401,83]	[443,89]	43.485	BRANCH	BRANCH
50	[401,83]	[394,103]	21.899	BRANCH	BRANCH
51	[443,89]	[436,136]	48.899	BRANCH	BRANCH
52	[389,103]	[394,103]	4.0	BRANCH	BRANCH
53	[389,103]	[356,125]	41.284	BRANCH	BRANCH
54	[394,103]	[430,136]	50.426	BRANCH	BRANCH
55	[276,98]	[272,110]	12.657	TIP	BRANCH
56	[267,113]	[272,110]	5.243	TIP	BRANCH
57	[274,111]	[272,110]	1.0	TIP	BRANCH
58	[7,117]	[17,117]	9.0	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
59	[0,90]	[7,117]	28.485	TIP	BRANCH
60	[3,125]	[7,117]	8.243	TIP	BRANCH
61	[17,117]	[52,143]	44.941	BRANCH	BRANCH
62	[502,117]	[488,119]	13.828	TIP	BRANCH
63	[488,119]	[468,146]	33.87	BRANCH	BRANCH
64	[356,125]	[324,128]	33.899	BRANCH	BRANCH
65	[356,125]	[354,141]	15.828	BRANCH	BRANCH
66	[304,116]	[324,128]	23.556	TIP	BRANCH
67	[324,128]	[320,152]	25.899	BRANCH	BRANCH
68	[430,136]	[436,136]	5.0	BRANCH	BRANCH
69	[430,136]	[419,150]	17.142	BRANCH	BRANCH
70	[436,136]	[468,146]	34.728	BRANCH	BRANCH
71	[354,141]	[351,170]	31.485	BRANCH	BRANCH
72	[354,141]	[320,152]	37.556	BRANCH	BRANCH
73	[91,110]	[52,143]	64.598	TIP	BRANCH
74	[52,143]	[56,168]	28.728	BRANCH	BRANCH
75	[468,146]	[468,148]	1.0	BRANCH	BRANCH
76	[468,148]	[467,163]	14.0	BRANCH	BRANCH
77	[468,148]	[479,157]	13.314	BRANCH	BRANCH
78	[419,150]	[401,164]	25.728	BRANCH	BRANCH
79	[419,150]	[438,182]	38.456	BRANCH	BRANCH
80	[320,152]	[312,158]	9.071	BRANCH	BRANCH
81	[504,146]	[479,157]	28.142	TIP	BRANCH
82	[504,170]	[479,157]	29.799	TIP	BRANCH
83	[199,177]	[312,158]	135.782	TIP	BRANCH
84	[312,158]	[314,215]	67.184	BRANCH	BRANCH
85	[467,163]	[438,182]	35.87	BRANCH	BRANCH
86	[467,163]	[486,234]	97.154	BRANCH	BRANCH
87	[401,164]	[379,181]	31.142	BRANCH	BRANCH
88	[401,164]	[434,187]	42.87	BRANCH	BRANCH
89	[74,183]	[56,168]	29.071	TIP	BRANCH
90	[66,163]	[56,168]	10.657	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
91	[351,170]	[327,213]	54.284	BRANCH	BRANCH
92	[351,170]	[367,186]	21.799	BRANCH	BRANCH
93	[379,181]	[367,186]	13.071	BRANCH	BRANCH
94	[379,181]	[396,207]	31.627	BRANCH	BRANCH
95	[438,182]	[434,187]	6.243	BRANCH	BRANCH
96	[367,186]	[367,204]	17.0	BRANCH	BRANCH
97	[434,187]	[427,215]	29.899	BRANCH	BRANCH
98	[24,146]	[8,203]	62.627	TIP	BRANCH
99	[0,221]	[8,203]	20.899	TIP	BRANCH
100	[15,210]	[8,203]	8.485	TIP	BRANCH
101	[367,204]	[341,217]	30.385	BRANCH	BRANCH
102	[367,204]	[373,210]	7.071	BRANCH	BRANCH
103	[396,207]	[401,210]	5.243	BRANCH	BRANCH
104	[396,207]	[373,210]	23.243	BRANCH	BRANCH
105	[373,210]	[375,232]	21.828	BRANCH	BRANCH
106	[401,210]	[427,215]	27.071	BRANCH	BRANCH
107	[401,210]	[375,232]	34.87	BRANCH	BRANCH
108	[327,213]	[314,215]	12.828	BRANCH	BRANCH
109	[327,213]	[341,217]	14.243	BRANCH	BRANCH
110	[314,215]	[304,229]	16.728	BRANCH	BRANCH
111	[427,215]	[427,235]	19.0	BRANCH	BRANCH
112	[339,237]	[341,217]	20.657	TIP	BRANCH
113	[304,229]	[275,236]	30.899	BRANCH	BRANCH
114	[304,229]	[299,267]	39.071	BRANCH	BRANCH
115	[375,232]	[368,246]	15.899	BRANCH	BRANCH
116	[465,220]	[477,234]	17.971	TIP	BRANCH
117	[477,234]	[486,234]	8.0	BRANCH	BRANCH
118	[477,234]	[451,239]	26.657	BRANCH	BRANCH
119	[486,234]	[497,283]	59.355	BRANCH	BRANCH
120	[427,235]	[451,239]	24.657	BRANCH	BRANCH
121	[427,235]	[402,281]	59.941	BRANCH	BRANCH
122	[217,222]	[275,236]	65.284	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
123	[275,236]	[275,249]	12.0	BRANCH	BRANCH
124	[451,239]	[453,265]	25.828	BRANCH	BRANCH
125	[368,246]	[333,260]	40.627	BRANCH	BRANCH
126	[368,246]	[369,271]	24.414	BRANCH	BRANCH
127	[275,249]	[255,263]	27.728	BRANCH	BRANCH
128	[275,249]	[276,269]	19.414	BRANCH	BRANCH
129	[333,247]	[333,260]	14.243	TIP	BRANCH
130	[333,260]	[331,266]	5.828	BRANCH	BRANCH
131	[251,265]	[255,263]	3.414	TIP	BRANCH
132	[255,263]	[275,272]	22.314	BRANCH	BRANCH
133	[453,265]	[478,282]	31.042	BRANCH	BRANCH
134	[453,265]	[432,287]	30.87	BRANCH	BRANCH
135	[331,266]	[299,267]	31.0	BRANCH	BRANCH
136	[331,266]	[334,286]	19.828	BRANCH	BRANCH
137	[299,267]	[285,283]	20.385	BRANCH	BRANCH
138	[278,269]	[276,269]	1.0	TIP	BRANCH
139	[276,269]	[275,272]	2.414	BRANCH	BRANCH
140	[385,266]	[384,269]	2.414	TIP	BRANCH
141	[384,269]	[402,281]	21.971	BRANCH	BRANCH
142	[384,269]	[369,271]	15.657	BRANCH	BRANCH
143	[369,271]	[360,332]	66.385	BRANCH	BRANCH
144	[275,272]	[278,284]	12.243	BRANCH	BRANCH
145	[402,281]	[403,284]	2.414	BRANCH	BRANCH
146	[478,282]	[497,283]	18.414	BRANCH	BRANCH
147	[478,282]	[453,301]	33.799	BRANCH	BRANCH
148	[285,283]	[278,284]	6.414	BRANCH	BRANCH
149	[294,291]	[285,283]	11.899	TIP	BRANCH
150	[497,283]	[497,309]	25.0	BRANCH	BRANCH
151	[278,284]	[271,306]	23.485	BRANCH	BRANCH
152	[403,284]	[432,287]	29.243	BRANCH	BRANCH
153	[403,284]	[399,307]	23.243	BRANCH	BRANCH
154	[334,286]	[318,303]	53.456	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
155	[334,286]	[318,303]	24.971	BRANCH	BRANCH
156	[111,275]	[109,287]	11.828	TIP	BRANCH
157	[102,295]	[109,287]	10.485	TIP	BRANCH
158	[121,290]	[109,287]	11.828	TIP	BRANCH
159	[432,287]	[453,301]	26.556	BRANCH	BRANCH
160	[453,301]	[452,313]	11.414	BRANCH	BRANCH
161	[311,306]	[318,303]	6.828	TIP	BRANCH
162	[93,273]	[76,304]	37.627	TIP	BRANCH
163	[100,299]	[76,304]	28.485	TIP	BRANCH
164	[76,304]	[35,345]	62.426	BRANCH	BRANCH
165	[271,306]	[240,310]	31.657	BRANCH	BRANCH
166	[271,306]	[296,322]	33.556	BRANCH	BRANCH
167	[399,307]	[360,332]	48.941	BRANCH	BRANCH
168	[399,307]	[430,326]	39.213	BRANCH	BRANCH
169	[507,310]	[497,309]	9.414	TIP	BRANCH
170	[497,309]	[486,331]	25.556	BRANCH	BRANCH
171	[240,310]	[213,320]	30.728	BRANCH	BRANCH
172	[227,279]	[240,310]	36.728	TIP	BRANCH
173	[452,313]	[430,326]	25.971	BRANCH	BRANCH
174	[452,313]	[467,328]	22.728	BRANCH	BRANCH
175	[203,295]	[213,320]	27.728	TIP	BRANCH
176	[160,325]	[213,320]	54.485	TIP	BRANCH
177	[299,321]	[296,322]	2.414	TIP	BRANCH
178	[292,331]	[296,322]	10.243	TIP	BRANCH
179	[430,326]	[439,341]	19.485	BRANCH	BRANCH
180	[467,328]	[486,331]	19.243	BRANCH	BRANCH
181	[467,328]	[439,341]	31.971	BRANCH	BRANCH
182	[486,331]	[487,338]	6.414	BRANCH	BRANCH
183	[360,332]	[358,339]	7.414	BRANCH	BRANCH
184	[510,359]	[487,338]	37.728	TIP	BRANCH
185	[487,338]	[478,379]	43.314	BRANCH	BRANCH
186	[358,339]	[336,341]	23.657	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
187	[358,339]	[336,341]	28.385	BRANCH	BRANCH
188	[358,339]	[369,358]	22.142	BRANCH	BRANCH
189	[336,341]	[271,364]	73.527	BRANCH	BRANCH
190	[449,373]	[439,341]	35.971	TIP	BRANCH
191	[32,271]	[35,345]	76.314	TIP	BRANCH
192	[1,360]	[35,345]	39.385	TIP	BRANCH
193	[159,350]	[189,357]	31.899	TIP	BRANCH
194	[189,357]	[222,382]	51.142	BRANCH	BRANCH
195	[201,348]	[189,357]	14.899	TIP	BRANCH
196	[396,339]	[369,358]	42.213	TIP	BRANCH
197	[369,358]	[369,364]	5.0	BRANCH	BRANCH
198	[270,360]	[271,364]	3.414	TIP	BRANCH
199	[255,394]	[271,364]	38.385	TIP	BRANCH
200	[369,364]	[358,374]	14.728	BRANCH	BRANCH
201	[369,364]	[391,384]	30.042	BRANCH	BRANCH
202	[332,366]	[329,374]	8.243	TIP	BRANCH
203	[329,374]	[358,374]	28.828	BRANCH	BRANCH
204	[304,400]	[329,374]	37.698	TIP	BRANCH
205	[361,384]	[358,374]	10.243	TIP	BRANCH
206	[478,379]	[503,412]	47.627	BRANCH	BRANCH
207	[478,379]	[462,393]	21.556	BRANCH	BRANCH
208	[222,382]	[278,434]	105.284	BRANCH	BRANCH
209	[248,339]	[222,382]	52.941	TIP	BRANCH
210	[391,384]	[414,386]	23.657	BRANCH	BRANCH
211	[391,384]	[391,393]	8.0	BRANCH	BRANCH
212	[426,377]	[414,386]	14.728	TIP	BRANCH
213	[414,386]	[408,408]	23.485	BRANCH	BRANCH
214	[391,393]	[383,420]	34.627	BRANCH	BRANCH
215	[391,393]	[408,408]	21.799	BRANCH	BRANCH
216	[462,393]	[438,412]	32.799	BRANCH	BRANCH
217	[462,393]	[503,412]	62.042	BRANCH	BRANCH
218	[408,408]	[408,412]	3.0	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
219	[408,412]	[438,412]	29.828	BRANCH	BRANCH
220	[408,412]	[383,420]	26.899	BRANCH	BRANCH
221	[438,412]	[441,428]	16.243	BRANCH	BRANCH
222	[511,426]	[503,412]	16.899	TIP	BRANCH
223	[383,420]	[382,422]	1.0	BRANCH	BRANCH
224	[382,422]	[336,427]	47.071	BRANCH	BRANCH
225	[391,445]	[382,422]	25.728	TIP	BRANCH
226	[336,427]	[336,429]	1.0	BRANCH	BRANCH
227	[326,421]	[336,427]	12.485	TIP	BRANCH
228	[441,428]	[496,487]	96.012	BRANCH	BRANCH
229	[434,432]	[441,428]	7.657	TIP	BRANCH
230	[336,429]	[278,434]	59.071	BRANCH	BRANCH
231	[336,429]	[354,474]	51.456	BRANCH	BRANCH
232	[267,407]	[278,434]	30.142	TIP	BRANCH
233	[354,474]	[396,481]	44.728	BRANCH	BRANCH
234	[356,509]	[354,474]	37.728	TIP	BRANCH
235	[401,464]	[399,477]	12.828	TIP	BRANCH
236	[405,476]	[399,477]	5.414	TIP	BRANCH
237	[399,477]	[396,481]	3.828	BRANCH	BRANCH
238	[415,492]	[396,481]	61.527	TIP	BRANCH
239	[0,389]	[19,484]	117.912	TIP	BRANCH
240	[17,492]	[19,484]	7.828	TIP	BRANCH
241	[0,471]	[19,484]	23.556	TIP	BRANCH
242	[506,508]	[496,487]	27.899	TIP	BRANCH
243	[496,487]	[458,504]	45.213	BRANCH	BRANCH
244	[427,468]	[458,504]	50.184	TIP	BRANCH
245	[449,510]	[458,504]	10.071	TIP	BRANCH
246	[200,1]	[206,15]	17.314	TIP	TIP
247	[190,29]	[192,29]	2.828	TIP	TIP
248	[190,65]	[159,78]	38.385	TIP	TIP
249	[299,74]	[314,74]	15.828	TIP	TIP
250	[241,120]	[220,130]	25.142	TIP	TIP

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
251	[162,156]	[128,160]	42.87	TIP	TIP
252	[124,168]	[97,171]	29.899	TIP	TIP
253	[87,191]	[102,203]	20.556	TIP	TIP
254	[225,192]	[190,223]	49.598	TIP	TIP
255	[265,199]	[271,205]	9.071	TIP	TIP
256	[56,202]	[64,223]	24.314	TIP	TIP
257	[69,238]	[68,273]	37.071	TIP	TIP
258	[27,243]	[34,246]	8.243	TIP	TIP
259	[180,254]	[175,272]	20.071	TIP	TIP
260	[144,263]	[130,279]	21.799	TIP	TIP
261	[10,309]	[2,313]	9.657	TIP	TIP
262	[149,337]	[120,353]	37.042	TIP	TIP
263	[38,361]	[57,399]	46.456	TIP	TIP
264	[177,397]	[167,405]	13.899	TIP	TIP
265	[311,445]	[300,468]	28.142	TIP	TIP
266	[175,450]	[171,453]	5.243	TIP	TIP
267	[251,470]	[270,497]	35.456	TIP	TIP
268	[158,475]	[159,492]	19.071	TIP	TIP
269	[235,475]	[244,496]	24.728	TIP	TIP
270	[194,485]	[197,485]	3.828	TIP	TIP
271	[56,503]	[52,510]	9.243	TIP	TIP